

FlowState

A Benchmark for Deployment Capacity and Alpha Decay in Indian Equities

measured under the information that is actually observable, and a negative result on what is not

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The benchmark in eight numbers

Rs. 0.81–7.91 cr	deployable capital for standard factors, set by the least liquid position	3 / 4	horizons where two defensible estimators disagree on the <i>sign</i> of impact
0 / 5	factors with a measurable decay half-life	0.836	split-half stability of Amihud illiquidity: the control that works
0.94–1.24×	capacity ratio between foreign outflow and inflow states; the predicted collapse is absent	0.374	split-half stability of the impact exponent, the parameter that does not
2 / 208,042	observations inside the order-size region of interest	125	strategies now carrying an audit verdict, a fragility score and a capacity

Benchmark, artifacts, corrections and evaluation protocol:
github.com/Ashwin-aadi/Trivijaya-Quant-Research

Abstract

The first question a portfolio manager asks about a strategy is how much money it can take. The standard answer is a capacity curve, and building one requires knowing how far a given order moves the price. **We measure whether daily bars can supply that knowledge, and report that they cannot—not for want of precision, but because the quantity is not identified.** Two equally defensible weighting schemes applied to the same reversal regression disagree about the *sign* of transient impact at 3 of 4 horizons, and nothing in the data adjudicates between them. Amihud illiquidity, computed on the same panel as a control, is stable across a sample split at 0.836 while the fitted impact exponent reaches only 0.374: the failure is in the model, not in the data.

We therefore release FlowState, a frozen benchmark that reports what daily bars do support. Deployment capacity is redefined as a *constraint*—the largest AUM at which a strategy can place every one of its trades inside a stated participation limit—which

is arithmetic on observed turnover and assumes no impact coefficient. On a survivorship-free Indian liquidity universe of 184 symbols over 2020–2024, standard price-and-volume factors deploy Rs.0.81–7.91 crore, and in every case the binding constraint is the least liquid position the strategy holds rather than the average liquidity of its book.

Two further results are null. **No factor’s edge decays measurably** across horizons from 1 to 63 sessions, and none is statistically established at the shortest horizon, so the precondition for a decay study fails. **Deployable size does not collapse when foreign participation reverses**—the effect this benchmark was designed to detect—with outflow-to-inflow ratios between 0.94 and 1.24. That null rests on a derivatives-based flow proxy and cannot distinguish a stable capacity from an uninformative proxy; we state this in the results rather than the discussion.

Finally, a result about benchmarks rather than about markets. Applying the finished benchmark to 156 machine-generated strategies *before* publishing it exposed two defects that 5 standard factor strategies had not, and repairing them moved every capacity figure *downward* by between a factor of four and a factor of twenty-six.

Contributions

1. **A measured negative result on identification.** Daily OHLCV cannot pin down a transient-impact coefficient. We demonstrate this four ways—weighting-dependence of the sign, a resolution argument, an unstable exponent, and an extrapolation gap—each with a detectability bound so that absence of evidence is separated from evidence of absence, and each against a control measure that does remain stable.
2. **A capacity definition that survives the negative result.** Constraint-based deployment capacity requires no impact model, is what a pre-trade risk system actually enforces, and is a strictly smaller claim than the one the data cannot support.
3. **Deployment capacities for standard factors on Indian equities**, which we believe are not openly published, together with the mechanism that sets them.
4. **A null on flow-conditional capacity**, reported with the limitation of its proxy in the same breath as the finding.
5. **A methodological result on benchmark validation:** a benchmark validated on a narrow, familiar strategy family failed to expose defects that a broader machine-generated population exposed immediately. We adopt the resulting ordering as standing process.
6. **A frozen, reproducible artifact** in which no published figure is transcribed. Every number in this paper resolves through a macro generated from an artifact, and a check fails the build if a bare numeral appears in a claim position.

1 Why capacity needs a benchmark

Every strategy has a size at which it stops working. Run a hundred crore through a mid-cap reversal signal and the signal is intact; run a thousand and the buying required to establish the position is itself the reason the position is unprofitable. The number at which that transition happens is the single most consequential fact about a strategy that a backtest does not report, and it is routinely either omitted or asserted.

The standard construction is a capacity curve: net return as a function of deployed AUM, declining as the strategy’s own trading pushes prices against it, with capacity read off where the curve crosses zero. Every point on that curve is a claim about market impact—how far a given order moves the price—and impact is not observable in a daily bar. It is therefore modelled, usually with the square-root law (Almgren et al., 2005), whose functional form is

standard and whose coefficient is calibrated on proprietary records of institutional orders in which the order and the price path it caused are separately visible.

We have no such record, and neither does anyone working from public Indian data. The question this paper opens with is whether the coefficient can be recovered anyway.

It cannot, and Section 3.4 is the measurement rather than the assertion. The obstruction is not noise. It is that a daily bar aggregates every participant's trading and cannot isolate one order, and—more decisively—that temporary impact decays over minutes to hours, so the finest resolution a daily bar offers is already coarser than the phenomenon being estimated.

1.1 What this paper is not

It is not a capacity curve. We do not report the AUM at which a strategy's edge reaches zero, because that quantity is not identified from the data available to us, and reporting it would require an impact coefficient we would have had to invent. A great many published backtests price market impact from daily bars; this paper is in part a measurement of what that assumption is worth.

It is not a claim that impact does not exist, or that it cannot be measured. It is a claim about what *daily bars* can identify. Intraday data would change the answer, and we report in Section 4 what we could and could not obtain.

It is not a study of Indian market microstructure. It is a benchmark: a frozen, versioned instrument that returns the same verdict on the same strategy tomorrow, so that different strategy generators can be compared against a fixed standard.

2 The benchmark

2.1 Universe, window, and what is held out

The universe is rules-based rather than index-based. At each quarterly rebalance it is the top names by trailing median traded value, computed only from sessions strictly prior to that date. Nothing consults a published membership list, which matters because a universe assembled from today's index constituents and applied backwards is survivorship-biased by construction: firms that were dropped after performing badly simply never appear. Under a rules-based definition a firm that deteriorates stops qualifying at the next rebalance and its bad period stays in the record.

Over 2020–2024 this yields 184 symbols and 122,464 investable symbol-days. A holdout year is reserved and **was never opened by any part of this project**.

2.2 The factor zoo, and the two families missing from it

Five signals, all constructible from price and volume alone: twelve-month momentum skipping the most recent month (Jegadeesh & Titman, 1993), one-month reversal, low realised volatility, Amihud illiquidity (Amihud, 2002), and a traded-value size proxy. Books are equal-weighted quintile long-short portfolios, formed every session, rupee-neutral by construction so that capacity scales linearly in deployed AUM.

Every signal is point-in-time by construction rather than by convention: each uses a trailing window ending strictly before the session it labels, enforced by shifting before rolling. A test truncates the price panel and re-derives every score, requiring bit-identical values for every session at or before the cut. A factor that peeks at its own future is indistinguishable from a good factor until somebody checks.

Value and quality are absent, and their absence is a scope limitation rather than an oversight. Both require company fundamentals—book value, earnings, accruals, return on equity—which this repository does not hold and could not acquire under a market-data budget fixed at zero. Any comparison against published factor results must be read knowing that the two most fundamentals-dependent families are missing. For the same reason the fifth factor is named `liquidity_size_proxy` rather than `size`: it is built from trailing traded value, not market capitalisation, which needs a share count we do not have. The earlier name invited a comparison against published size-factor results that would not have been like-for-like.

2.3 Deployment capacity, defined

For a book rebalancing from weights w to w' , the traded fraction in name i is $|\Delta w_i|$. At deployed AUM A the rupee trade is $A|\Delta w_i|$, and the participation constraint permits at most κ of that session's traded value $V_i(t)$. Hence

$$A_{\max}(t) = \min_{i: |\Delta w_i| > \varepsilon} \frac{\kappa \cdot V_i(t)}{|\Delta w_i(t)|}, \quad A_{\max} = \min_t A_{\max}(t).$$

Three details in that expression are load-bearing.

$V_i(t)$ is a trailing average, not the session's own volume. Sizing an order against the volume that the order itself helped create is circular, and the circularity flatters capacity.

The outer minimum runs over sessions, not a median. A strategy must be executable on every rebalance it makes, not on a typical one. This also makes the session that builds the opening position binding for a near-static strategy, so initial portfolio construction is priced like any other trade rather than treated as free. Section 3.5 describes what a median concealed before we corrected it.

ε excludes weight changes too small to be trades. A strategy that recomputes an unchanged weight produces differences of order 10^{-18} , and dividing a participation limit by that residue produces a capacity of 10^{23} crore. We set $\varepsilon = 1.0 \times 10^{-9}$ of the book.

2.4 Reproducibility, and how every number here was produced

No number in this paper was typed. Each figure is computed by a generator from a frozen artifact, emitted as a macro, and referenced by name; a companion check parses the paper, strips structural markup, and fails the build if any bare numeral survives in a claim position. Numeric literals that are genuinely not results—a holding horizon, the power of a statistical test—appear on a short allowlist, each with a written reason. A number that is genuinely a measurement cannot be added to that list honestly.

The gate is tested by breaking it. A test takes the committed paper, substitutes one macro for the value it renders to—exactly as a careless edit would—and requires the check to start failing. A gate nobody has seen fail is indistinguishable from a gate that cannot fail, so the control runs on every commit rather than existing as a one-time manual act.

Each paper in this programme owns a macro namespace, enforced by a further test, so that a figure generated for one paper can never silently satisfy a reference in another. Two papers would then share a number without sharing an artifact, which is the precise failure the apparatus exists to prevent. The global seed is fixed by the project charter; every script writes a run manifest recording git SHA, configuration hash, package versions and input file hashes.

3 Results

3.1 Deployment capacity, and what sets it

Factor	Capacity	Entry	Median session	Med ÷ bind	One name
momentum_12_1	6.94	28.61	31.4	4.5	27.4
reversal_1m	0.81	9.89	21.2	26.2	7.7
low_volatility	6.78	8.31	32.9	4.8	40.1
illiquidity	7.22	10.75	32.0	4.4	24.1
liquidity_size_proxy	7.91	16.02	33.9	4.3	18.8

Table 1. Deployment capacity in crore of rupees at a 1.0% participation limit. *Capacity* is the binding session; *entry* is the session that builds the opening position; the median session and its ratio to the binding figure show what a typical-session summary would have reported instead. *One name* is the share of sessions on which the tightest position is more than twice as tight as the next.

Deployment capacity is **Rs. 0.81–7.91 crore**, roughly one to ten million US dollars. **The least liquid position the strategy holds decides it**, not the average liquidity of its book.

The mechanism, because the number is otherwise easy to misread. These are equal-weighted books. Every name in a leg carries the same weight and must therefore be traded in the same rupee size. A strategy holding a name that turns over a tenth of what another turns over trades both identically, and the participation constraint binds on the thinner one first. Capacity is set there, for the whole book.

The consequence is counter-intuitive and worth stating plainly: **a strategy holding a hundred large-capitalisation names can still be capacity-constrained by one of them.** Diversification across many liquid names does not raise capacity if the weighting scheme forces the thinnest name to be traded at the same size as the deepest. The *one name* column reports how often this is literally a single position.

This is not an artefact of our definition—it is the constraint a desk actually faces—but it is a property of *equal weighting* as much as of the market. The standard remedies, liquidity-tilted weights or a liquidity screen at entry, both change the portfolio and therefore change the returns that the capacity is a capacity *for*. We measure the strategies as specified rather than silently repairing them, and note that a capacity figure is only meaningful when paired with the weighting scheme that produced it.

Dropping the single tightest position each session raises capacity by well under an order of magnitude. That robustness check is what establishes these figures describe the strategy rather than one holding; had relaxation bought tenfold, the headline would have been a statement about one name.

3.2 Alpha decay

0 of 5 factors have a measurable half-life. Every curve is flat to gently rising out to a quarter.

We report this as a null about our measurement rather than as a finding about Indian markets, because the precondition fails. The largest $|t|$ at the shortest horizon is 1.79, and the

Factor	Return per session held (bps)				<i>t</i> -statistic			
	<i>h</i> =1	<i>h</i> =5	<i>h</i> =21	<i>h</i> =63	<i>h</i> =1	<i>h</i> =5	<i>h</i> =21	<i>h</i> =63
illiquidity	+4.86	+4.96	+5.62	+6.82	+1.79	+1.75	+2.07	+2.40
momentum_12_1	+3.96	+3.83	+5.15	+5.94	+1.12	+1.08	+1.50	+1.95
liq_size_proxy	+2.57	+2.44	+3.02	+3.88	+1.22	+1.14	+1.49	+2.12
low_volatility	−3.02	−3.79	−4.92	−5.33	−0.73	−0.93	−1.28	−1.39
reversal_1m	−6.22	−6.59	−3.40	−1.74	−1.79	−2.04	−1.05	−0.50

Table 2. Long-short return per session held, gross of costs, across 1,233 formation dates. Books are formed every session and held for *h* sessions after a one-session execution gap, so consecutive observations overlap; *t*-statistics divide by the number of independent holding periods rather than the raw count, which is deliberately conservative.

largest at any horizon in the table is 2.40. **The edges whose decay is being measured are not themselves established**, and a decay half-life is only a meaningful quantity for a signal that had an edge to lose. Reporting one for a factor that never made money would be a category error dressed as a measurement.

Two factors carry the wrong sign for their names, which is itself informative about the window. **reversal_1m** loses money, so at a one-month horizon this period shows *continuation* rather than reversal. **low_volatility** loses, so high-beta names outperformed—consistent with a window containing the COVID drawdown and the recovery that followed it, in which the names that fell hardest rebounded hardest.

A research question retired rather than deferred. This benchmark was specified to ask whether alpha decays faster in India than in developed markets and whether retail participation explains it. The first clause is answered: no decay was observed. The second therefore loses its antecedent—there is nothing for retail participation to explain—and is retired. Delivery-share data that would have supported a retail analysis is freely available and was deliberately not collected, on the principle that the availability of data is not a reason to ask a question.

3.3 Flow-conditional capacity

Outflow-to-inflow capacity ratios lie between 0.94 and 1.24 across 5 factors, median 0.96, on 76–234 sessions per cell. **The hypothesis is not supported.**

The novel contribution this benchmark was built for was to ask whether deployable size collapses when foreign participation reverses. India is a plausible place to look: foreign flows are large, volatile, and widely believed to drive liquidity. A tenfold swing in capacity between inflow and outflow regimes would be a useful and previously unquantified fact.

It is not there. The ratios are within a quarter of unity and some point the wrong way.

This null rests on a proxy, and we cannot separate two explanations. No free source serves historical daily foreign and domestic institutional *cash-market* net flow. The exchange publishes the series but its endpoint returns the current session regardless of the date requested, which we verified against three parameter spellings; the alternative sources we tried returned error pages or refused the request. The series we use instead is foreign-institutional activity in *equity derivatives*, and it differs from the intended series in three ways that are not small. It is denominated in contracts rather than rupees, over a window in which regulatory lot

sizes changed. It describes derivatives rather than cash. And foreign derivatives activity is substantially hedging against cash positions, so its sign need not track the direction of cash-market demand at all.

Whether capacity is genuinely stable across flow states, or the proxy is simply uninformative about cash liquidity, is not separable with this data. We state this beside the result rather than in a limitations section, because it governs how the null may be read.

3.4 Why daily OHLCV cannot price transient impact

Horizon (sessions)	Precision-weighted β	Equal-weighted β	MDE pooled	MDE one symbol
1	-0.0180	-0.0114	0.0135	0.182
5	-0.0395	+0.0205	0.0267	0.368
10	-0.0588	+0.0370	0.0350	0.497
21	-0.0996	+0.0867	0.0477	0.699

Table 3. Reversal regressions (Campbell et al., 1993) of the following h -session return on today's, restricted to each symbol's heaviest-volume decile, $n = 177$ symbols. MDE is the minimum effect distinguishable from zero at 80% power. Precision weighting is inverse-variance pooling; equal weighting counts each symbol once.

A price move caused by an order is *transient*: it is the cost of demanding liquidity and it is repaid once the demand stops. A move caused by information is permanent. A negative reversal coefficient is therefore the signature of impact, and its sign is the quantity on which any daily-bar impact estimate depends.

The sign depends on the weighting scheme. Precision weighting and equal weighting are both standard, neither is wrong, and they **disagree on sign at 3 of 4 horizons**. A quantity whose sign is determined by an analyst's choice among defensible estimators is not identified by the data.

The disagreement is not a puzzle to be resolved by picking the better estimator; it is the result. Inverse-variance pooling is correct when symbols estimate a common parameter and differ only in noise. Equal weighting is correct when the parameter itself varies across symbols and the quantity of interest is its population average. **Nothing in the data tells us which world we are in**, and the two answers have opposite signs. Choosing between them requires prior knowledge of exactly the microstructure the exercise was supposed to measure.

Three further measurements agree, and one control locates the failure:

- **Resolution.** Temporary impact decays over minutes to hours; the finest horizon a daily bar permits is one session. The pooled coefficient also grows monotonically more negative as the horizon lengthens rather than decaying, which is the signature of slow mean reversion, not of a liquidity concession being repaid.
- **Exponent.** Fitting $|r|$ against relative traded value gives a median exponent of 0.689 against the square-root law's 0.5, spanning 0.521–0.859 across 179 symbols and explaining a median 0.126 of variance. Across a sample split the per-symbol exponent correlates 0.374 with itself ($n = 166$). A parameter that does not survive its own sample split is not a parameter.
- **Extrapolation.** A capacity study asks about 1.0% participation. Observed relative traded

value has a median of 0.821—1.91 orders of magnitude away—and only 2 of 208,042 symbol-days fall inside the region of interest. Fitting here and applying there is not interpolation.

- **The control.** Amihud illiquidity, computed on the same panel, correlates 0.836 across the same split ($n = 166$, $p = 1.7 \times 10^{-44}$). The data supports a stable liquidity *measure* while refusing to support an impact *model parameter*. **The failure is in the model, not in the data**—which is the whole reason the control is reported.

Two corrections to our own earlier reporting of this result. An earlier version reported that *no* reversal was detected, citing the proportion of symbols with a negative coefficient as evidence. That proportion was noise. A single symbol’s daily test can detect only an effect of 0.368 or larger at the five-session horizon—an order of magnitude above the pooled coefficient of -0.0395 that the same data supports—so the proportion was always going to land near half regardless of the truth. We had reported an underpowered statistic as a measurement.

Computing the detectability bound then showed that the pooled coefficient *is* significantly negative, contradicting our own earlier claim outright. The identification argument above replaced it and is stronger. We record the sequence because the null we first reported was not the null that survived, and a reader is entitled to know that the argument moved.

3.5 The validation gap

The benchmark was validated on 5 standard factor strategies spanning $9.8\times$ in capacity, then applied to 156 machine-generated strategies spanning $30.9\times$ — $3.2\times$ wider. **Two defects lived in that gap, and the corpus found both immediately.**

Before freezing the benchmark we ran it across a corpus of strategies generated by a local language model and previously evaluated by two companion benchmarks in the same programme. The intent was to produce a reference corpus in which every strategy carries an audit verdict, a fragility score and a deployment capacity; 125 strategies do. What it produced first was two bugs.

Floating-point residue counted as trading. A strategy holding a hundred names at equal weight and never rebalancing produces weight changes of order 10^{-18} from recomputing an unchanged number. Dividing a participation limit by that residue reported its capacity as 10^{23} crore. The remedy is the tolerance ε defined earlier.

The summary statistic discarded the binding constraint. Capacity was reported as the *median* session. For a near-static book that is close to meaningless: almost every session has near-zero rebalancing turnover, so the one tightly constrained session that builds the position was drowned by a thousand that barely trade. The remedy is the minimum over sessions.

A subtlety worth recording, because we diagnosed it wrongly at first. We initially believed the formula never counted the acquisition of the opening position. It did—a strategy’s first session enters at full weight from zero. **The defect was that the summary statistic discarded it**, which is a different error and took a direct check to see. Using the median would overstate the typical corpus strategy by $10.6\times$, and entry is the binding session for only 32 of 156 strategies (20.5%)—so the median was concealing considerably more than the entry constraint alone.

Neither defect was reachable from the validation set. All five factor strategies rebalance substantially every session, so none ever produced a near-zero traded fraction, and for all five the median and the minimum differ by a factor small enough to look like a modelling choice rather than a bug.

The corrections moved every published number down. Capacity figures fell by between a factor of four and a factor of twenty-six. We record the direction because it bears on whether this counts as tuning: adapting a benchmark to flatter its subject would not systematically reduce the results it reports. Repairing an objective omission in a definition is not the same act as adjusting a threshold until a result improves, and the sign of the change is the cheapest available evidence of which one happened.

The general form: **a benchmark's soundness cannot be established on the population that motivated it.** Defects concentrate in the tails of the population a benchmark is applied to, and a validation set assembled from familiar, well-behaved strategies has no tails.

We have adopted the resulting ordering as standing process for every benchmark in this programme: **build, validate on standard strategies, apply to the reference corpus, freeze, write up.** The corpus run precedes the freeze and the paper. Had the original ordering held, this paper would have shipped a capacity table containing 10^{23} crore.

4 Threats to validity

1. **The flow proxy.** The flow-conditional null rests on derivatives activity rather than cash-market flow. This is the threat we would attack first, and it is stated in the abstract and in the result rather than only here.
2. **Capacity is an extreme-value statistic.** The headline is a minimum over roughly twelve hundred sessions and is therefore sensitive to any one of them. We report the fifth percentile and a drop-the-tightest-name variant alongside it, but a minimum is a minimum and should be read as one.
3. **The factor zoo is incomplete.** Value and quality are absent for want of fundamentals, which weakens any comparison against published factor results and means the zoo over-represents liquidity-driven signals.
4. **Decay is measured on edges that are not established.** With no factor significant at the shortest horizon, the decay curves describe the decay of quantities that may be zero. A flat curve through zero is not evidence of persistence.
5. **Returns are gross of costs.** Indian transaction costs are large enough to reverse the sign of a small edge, and every decay figure here would fall under them.
6. **Equal weighting drives the capacity result.** The finding that the least liquid position binds is a joint property of the market and of the weighting scheme. A different scheme would give different capacities for the same signals.
7. **One market, one window.** 2020–2024 in Indian equities contains one pandemic drawdown and one recovery. Every negative result here may be a property of that window.

8. **The corrections were made after seeing the corpus.** Both defects in Section 3.5 were found by data the benchmark was about to be applied to, and repaired before freezing. We argue this repairs objective omissions rather than tuning, and note the corrections all moved results downward, but a reader is entitled to weigh that themselves.
9. **Intraday data would change the identification result.** We established that no free source reaches this window at intraday resolution, and that commercially available sources at a retail price point reach only part of it. With minute bars the extrapolation gap closes and the transience test can be run at the scale impact actually operates. Our claim is about daily bars, not about the phenomenon.

5 Conclusion

FlowState reports what daily bars can support and declines what they cannot. Deployment capacity for standard price-and-volume factors on an Indian liquidity universe is Rs. 0.81–7.91 crore, set in every case by the least liquid position each strategy holds rather than by the average depth of its book. No factor’s edge decays measurably within a quarter, and none is established at the shortest horizon. Deployable size does not collapse when foreign participation reverses, though that null is conditional on a proxy we would rather not have needed.

The result we expect to outlast the others is the one about instruments rather than about markets. Daily OHLCV cannot identify a transient-impact coefficient—not for want of precision, but because two equally defensible estimators disagree about its sign and the data cannot arbitrate. A great many published backtests price market impact from daily data. This is a measurement of what that assumption is worth, and it comes with a control showing that the same data supports a stable liquidity measure perfectly well.

FlowState is intended as benchmark infrastructure rather than a completed scientific result. It is frozen, versioned, and reproducible from raw data by a documented command sequence, so that future strategy generators, future capacity definitions and future impact estimators can be evaluated against the same protocol and progress can be measured against a common standard. The corpus it now carries—125 strategies with an audit verdict, a fragility score and a deployment capacity apiece—exists to make the next comparison possible.

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